

Rendering

The creation of a 3D scene involves a process known as rendering, in which a mathematical description of the scene is converted into a graphical scene to be displayed. In this process, the initial work is to determine the geometry limit that the graphics card can achieve. Once the geometry limit is determined, the amount of lighting, texture and colour of each individual pixel is calculated. So once this process is done, you get the complete scene ready to be displayed and please your eyes.

tions and makes the objects cleaner. This means that the higher the filtering, the cleaner the things will be.

The images created in advance by the GPU are stored in a part of the card's RAM that acts as the frame buffer.

This RAM even stores the information about each pixel's colour and location on the screen.

The card deploys a random-access memory digital-to-analogue converter (RAMDAC), which converts the digital output produced by the graphics processor to analogue required for monitors

using CRT displays and LCD displays that use analogue input. This RAMDAC supports different refresh rates due to change in the number of bits used and RAMDAC transfer-rate. The RAM is directly connected to the RAMDAC.

The card even provides video graphics array (VGA), digital visual interface (DVI), video-in video-out (VIVO) and high-definition multimedia interface (HDMI) outputs.

The PCIe interface for connection between the motherboard and the graphics card provides fastest transfer rates and supports connection of altogether two external graphics cards.

The main factors that determine the quality of a graphics card are its memory and frame rate. Frame rate accounts highly for the card's performance and is measured in frames per second. It describes how many complete images the graphics card will be able to display per second. The human eye and its brain interface, i.e., the human visual system, can process ten to twelve separate images per second,

perceiving them individually.

Frame rate even consists of triangles or vertices per second and pixel fill rate.

Triangles per second is a graphics industry performance indicator. Since nearly every render in modern computer graphics interface is made up of triangles, the higher the number of triangles per second, the faster the graphics processing unit is rated. Nvidia GTX 580 can process about two billion per second.

Faster graphics processing units mean more complex renderings in real time, which greatly improves immersion.

Pixel fill rate indicates the number of pixels the card can render to the screen every second when redrawing a 3D scene on a video display. It is calculated by multiplying the number of horizontal lines of pixels by the clock speed of the graphics hardware but varies according to the complexity of the scene to be displayed. ●

The authors are BE third year students at NMAMIT, Nitte

India's #1 Website For Electronics Engineers Working Across The Globe

Established: 1998 | Registered Users: 250,000+

Monthly Statistics

Page Impressions: 750,000+
Unique Techies: 150,000

BETA
electronicsforu.com

If it's electronics, it's here



For any queries, please contact our team at diyeng@efyindia.com OR +91-11-26810601